

Retail Sales & Profitability Analytics Dashboard for ElectroHub(case)

Company Overview-

ElectroHub is a retail company that deals with multiple product categories including Electronics, Clothing, Footwear, Home Appliances, Accessories, Kitchenware, Bags, and Personal Care products.

As the business expanded across categories and regions, it became difficult to track overall sales performance, profitability, product demand, and discount impact through traditional reporting methods. The company required a centralized analytics solution that could provide a clear view of business performance and help teams make better data driven decisions.

The goal of this project was to analyze transactional sales data and build an interactive BI dashboard that could track key business KPIs, identify product performance trends, monitor regional sales distribution, and evaluate the impact of discounts on overall profitability.

Business Requirements

The main objective of the project was to create an interactive analytics dashboard that could help ElectroHub monitor business performance across products, regions, and time periods.

The dashboard was designed to address the following business requirements:

- Identify the top and bottom performing products based on sales, profit, and quantity sold.
- Analyze sales trends across daily, monthly, quarterly, and yearly timelines.
- Understand the relationship between sales and profitability across different product categories.
- Compare sales, profit, and quantity sold between two different time periods selected by the user.

- Evaluate the average discount offered under different discount categories.
 - Track the total number of customer orders.
 - Provide detailed order-level reporting with dynamic filtering options using product, date, customer ID, and promotion categories.
 - Visualize sales distribution across different cities to identify high performing regions.
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Dataset Overview

The project was built using a retail sales dataset consisting of four interconnected tables representing customers, products, promotions, and sales transactions.

a) Customer Table

Contains customer related information used for customer level analysis and regional sales reporting.

Fields included:

- Customer ID
- Customer Name
- City
- State
- Pincode
- Email ID
- Phone Number

b) Product Table

Stores product details across different retail categories.

Fields included:

- Product ID
- Product Name
- Product Line
- Price (INR)

c) Promotion Table

Contains promotional campaign and discount related information used to analyze discount impact on sales and profitability.

Fields included:

- Promotion ID
- Promotion Name
- Ad Type
- Coupon Code
- Price Reduction Type

d) Sales Transactions Table

Acts as the central fact table containing transactional sales records and business metrics.

Fields included:

- Date
- Customer ID
- Promotion ID
- Product ID
- Units Sold
- Price Per Unit
- Total Sales
- Discount Percentage
- Discount Value
- Net Sales

Data Import & Test Environment Setup

The raw retail datasets were initially available in CSV format and were imported into Microsoft SQL Server for structured data handling and preprocessing.

To simulate a real world BI workflow, a separate testing environment named **ElectroHub_Test** was created before working on the complete production dataset.

The original sales transactions table contained nearly 3,000 records, out of which a smaller sample of approximately 300 rows was retained in the testing environment.

After initial validation, the test data was imported into Power BI Desktop for data modeling and dashboard development.

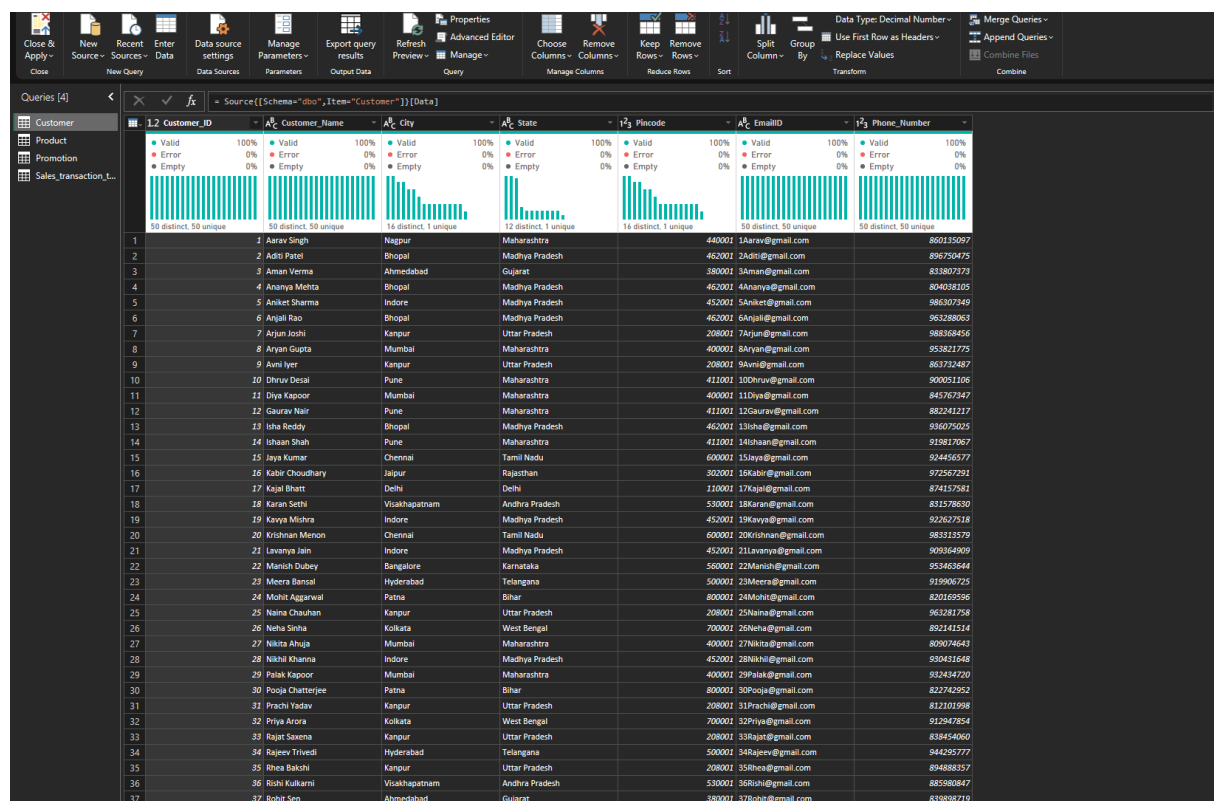
This staged approach helped improve reporting reliability and made the development process easier to manage during the initial phases of the project.

Data Cleaning & Transformation

Data preprocessing and transformation were performed using Power Query Editor in Power BI before building the final dashboard.

1. Data Type Validation

The first step was to review and validate the data types of all columns across the imported tables. This helped ensure that numerical calculations, date operations, and relationships worked correctly during analysis.



2. Extracting Discount Percentage

The **Price Reduction Type** column in the Promotion table contained discount information in text formats such as:

- 30% Off
- Buy 1 Get 1 Free

To make the data usable for calculations, a new conditional column named **Discount Percentage** was created using Power Query transformations. This step helped standardize discount-related information for further KPI calculations.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Custom

	Column Name	Operator	Value	Output
If	PromotionID	equals	PR001	10
Else If	PromotionID	equals	PR002	20
Else If	PromotionID	equals	PR003	50
Else If	PromotionID	equals	PR004	50
Else If	PromotionID	equals	PR005	60

Add Clause

Else

OK Cancel

3. Handling Missing Values in Sales Transactions

Several columns in the Sales Transactions table, including:

- Price Per Unit
- Total Sales
- Discount Percentage
- Discount Value
- Net Sales

initially contained null values and needed to be populated.

4. Populating Price Per Unit using Merge Operations

To populate the **Price Per Unit** field, a merge operation was performed between the Sales Transactions table and the Product table using **Product ID** as the common key.

The required pricing information was then expanded into the Sales Transactions table.

Merge

Select a table and matching columns to create a merged table.

Sales_transaction_test

Date_dd_mm_yyyy	CustomerID	PromotionID	Product_ID	Units_Sold	Total_Sales	Discount_Percentage
04-02-2020	10	0	4.00	2	null	
19-05-2022	28	PR001	27.00	3	null	
01-07-2022	2	0	11.00	3	null	
06-04-2022	11	PR001	9.00	2	null	

Product

ProductID	Product_Name	Product_Line	Price_INR
1.00	Apple iPhone 14	Electronics	79999
2.00	Samsung Galaxy S21	Electronics	69999
3.00	Sony Bravia 55" TV	Electronics	89999
4.00	HP Pavilion Laptop	Electronics	64999
5.00	Apple MacBook Air	Electronics	92499

Join Kind
Left Outer (all from first, matching from second)

☒ Use fuzzy matching to perform the merge

Fuzzy matching options

✓ The selection matches 299 of 3510 rows from the first table.

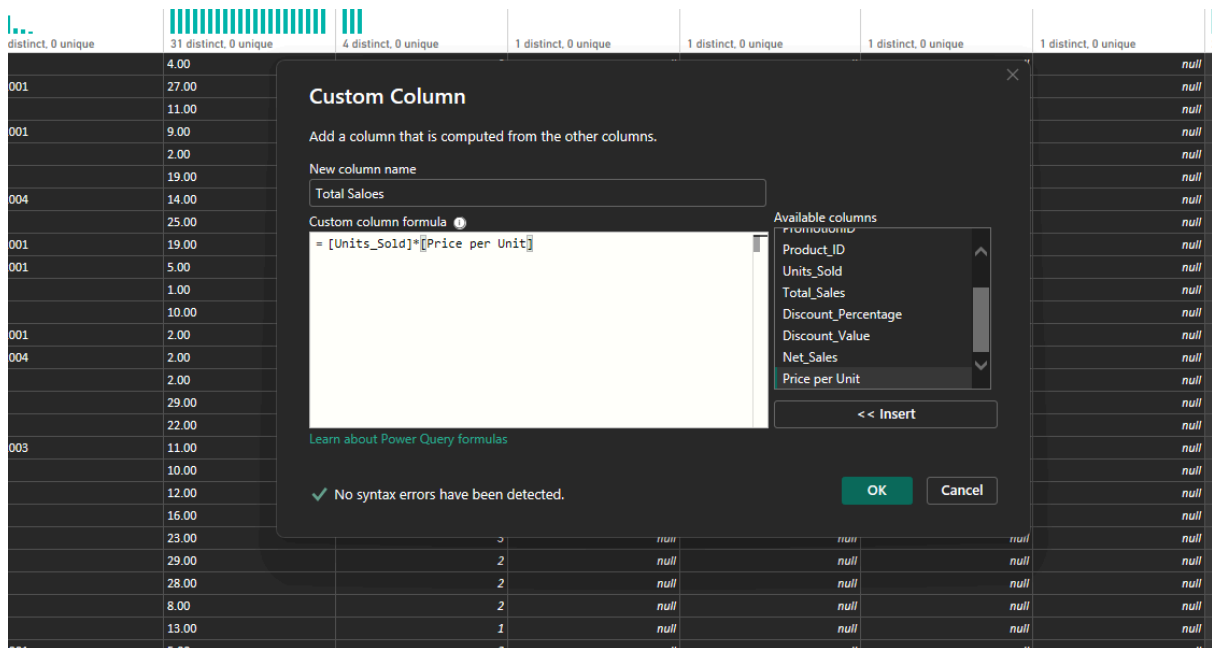
OK Cancel

5. Calculating Total Sales

A new calculated column named **Total Sales** was created using the formula:

Units Sold × Price Per Unit

This calculation helped generate the base revenue value for each transaction.



6. Handling Discount Values

The discount percentage values were populated into the Sales Transactions table from the Promotion table. For products where no discount information was available, null values were replaced with 0 to avoid calculation errors.

7. Calculating Discount Value

The **Discount Value** column was calculated using:

$(\text{Total Sales} \times \text{Discount Percentage}) / 100$

This represented the actual discount amount applied to each transaction.

8. Calculating Net Sales

The final **Net Sales** metric was calculated by subtracting the discount amount from the total sales value:

$\text{Net Sales} = \text{Total Sales} - \text{Discount Value}$

This metric was later used as one of the primary KPIs in the dashboard analysis.

Date_dd_mm_yyyy	CustomerID	PromotionID	Product_ID	Units_Sold	Price_Per_unit	Total Sales	Discount Percentage	Discount Value	Net Sales
18 April 2021	17	PR001	17.00	1	10499	10499	10	1049.9	9449.1
27 April 2020	38	PR001	17.00	3	10499	31497	10	3149.7	28347.3
29 December 2023	6	PR005	17.00	1	10499	10499	70	7349.3	3149.7
30 December 2023	46	PR005	18.00	1	2299	2299	70	1609.3	689.7
30 December 2023	13	PR005	18.00	2	2299	4598	70	3218.6	1379.4
11 July 2020	18	PR004	18.00	3	2299	6897	50	3448.5	3448.5
22 May 2021	11	PR001	18.00	2	2299	4598	10	459.8	4138.2
09 April 2023	13	PR001	18.00	2	2299	4598	10	459.8	4138.2
31 December 2023	29	PR005	18.00	1	2299	2299	70	1609.3	689.7
29 May 2022	4	PR001	18.00	3	2299	6897	10	689.7	6207.3
24 May 2021	11	PR001	18.00	1	2299	2299	10	229.9	2069.1
29 December 2023	33	PR005	19.00	1	5999	5999	70	4199.3	1799.7
17 May 2023	47	PR001	19.00	1	5999	5999	10	599.9	5399.1
31 December 2023	33	PR005	19.00	3	5999	17997	70	12597.9	5399.1
25 May 2022	21	PR001	19.00	2	5999	11998	10	1199.8	10798.2
20 April 2020	1	PR001	2.00	1	69999	69999	10	6999.9	62999.1
30 April 2022	38	PR001	2.00	3	69999	209997	10	20999.7	188997.3
29 May 2023	17	PR001	2.00	3	69999	209997	10	20999.7	188997.3
08 July 2023	28	PR004	2.00	3	69999	209997	50	104998.5	104998.5
29 December 2023	39	PR005	20.00	1	12999	12999	70	9099.3	3899.7
29 December 2023	24	PR005	21.00	2	1299	2598	70	1818.6	779.4
30 December 2023	36	PR005	22.00	2	1999	3998	70	2798.6	1199.4
29 May 2022	24	PR001	22.00	2	1999	3998	10	399.8	3598.2
30 April 2022	3	PR001	22.00	2	1999	3998	10	399.8	3598.2
30 December 2023	38	PR005	23.00	1	1499	1499	70	1049.3	449.7
15 April 2020	18	PR001	23.00	3	1499	4497	10	449.7	4047.3
07 May 2020	20	PR001	23.00	3	1499	4497	10	449.7	4047.3
04 July 2020	49	PR004	23.00	1	1499	1499	50	749.5	749.5
20 April 2021	13	PR001	24.00	3	2799	8397	10	839.7	7557.3
10 April 2020	12	PR001	24.00	2	2799	5598	10	559.8	5038.2
01 January 2022	32	PR003	24.00	3	2799	8397	50	4198.5	4198.5
10 May 2022	47	PR001	25.00	1	8999	8999	10	899.9	8099.1
09 April 2020	16	PR001	25.00	1	8999	8999	10	899.9	8099.1
06 April 2021	15	PR001	26.00	3	5499	16497	10	1649.7	14847.3
31 July 2022	27	PR004	26.00	2	5499	10998	50	5499	5499
01 May 2020	30	PR001	26.00	2	5499	10998	10	1099.8	9898.2
02 April 2023	32	PR001	26.00	2	5499	10998	10	1099.8	9898.2
15 May 2022	21	PR001	26.00	2	5499	10998	10	1099.8	9898.2
27 April 2022	48	PR001	27.00	1	699	699	10	69.9	629.1
19 May 2022	28	PR001	27.00	3	699	2097	10	209.7	1887.3
10 May 2022	47	PR001	28.00	2	349	698	10	69.8	628.2
22 May 2021	42	PR001	29.00	3	99	297	10	29.7	267.3
14 April 2023	14	PR001	29.00	2	99	198	10	19.8	178.2
22 July 2023	20	PR004	29.00	2	99	198	50	99	99
29 December 2023	36	PR005	30.00	3	399	1197	70	837.9	359.1

Data Modeling & Schema Design

After completing the data transformation process, the tables were connected in the Data Model view of Power BI to build a structured relational model for analysis.

A star schema approach was used, where the **Sales Transactions** table acted as the central fact table and the remaining tables were treated as dimension tables.

Fact Table

The **Sales Transactions** table was used as the fact table since it contained transactional business metrics such as:

- Units Sold
- Total Sales
- Discount Value
- Net Sales

This table served as the primary source for KPI calculations and reporting.

Dimension Tables

The following tables were connected as dimension tables:

- Customer
- Product
- Promotion

These tables provided descriptive information used for filtering, grouping, and drill-down analysis.

Relationships Created

Customer Relationship

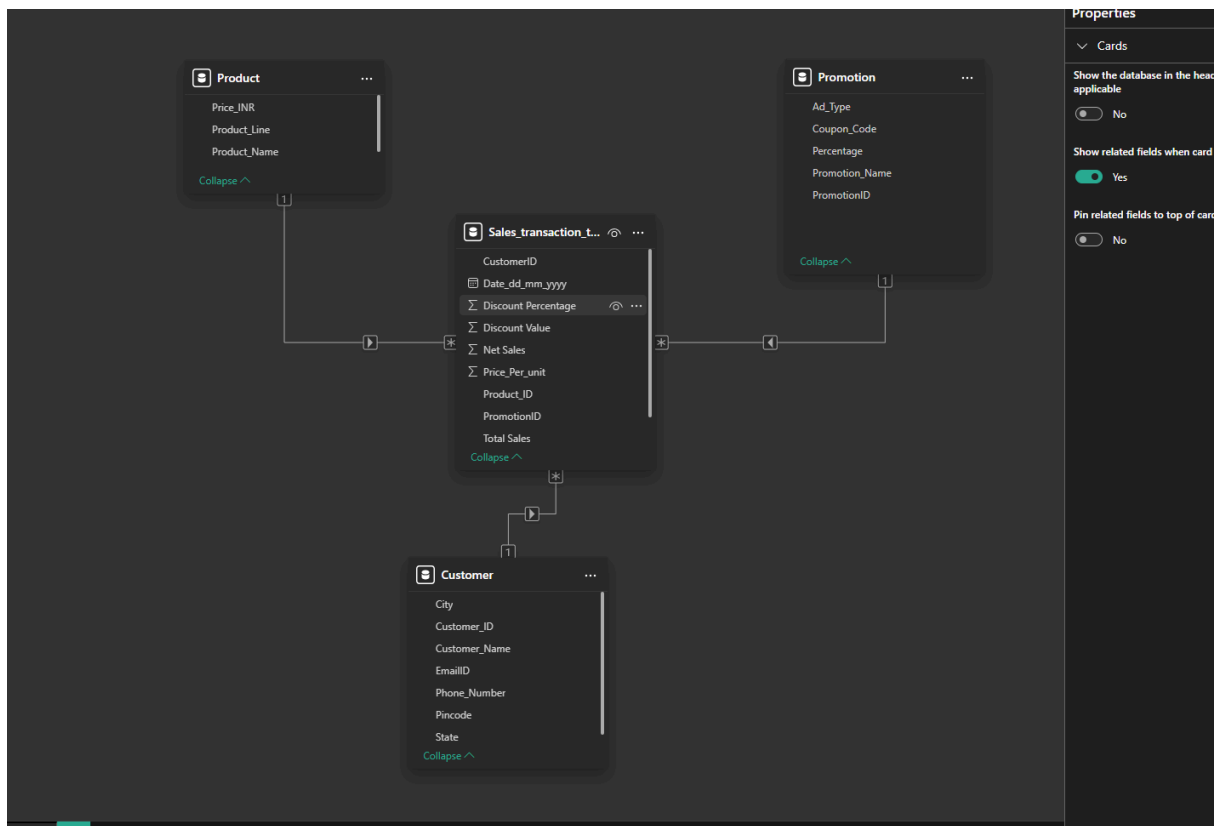
A relationship was created between:

- `Sales Transactions[Customer ID]`
- and
- `Customer[Customer ID]`

The relationship was configured as:

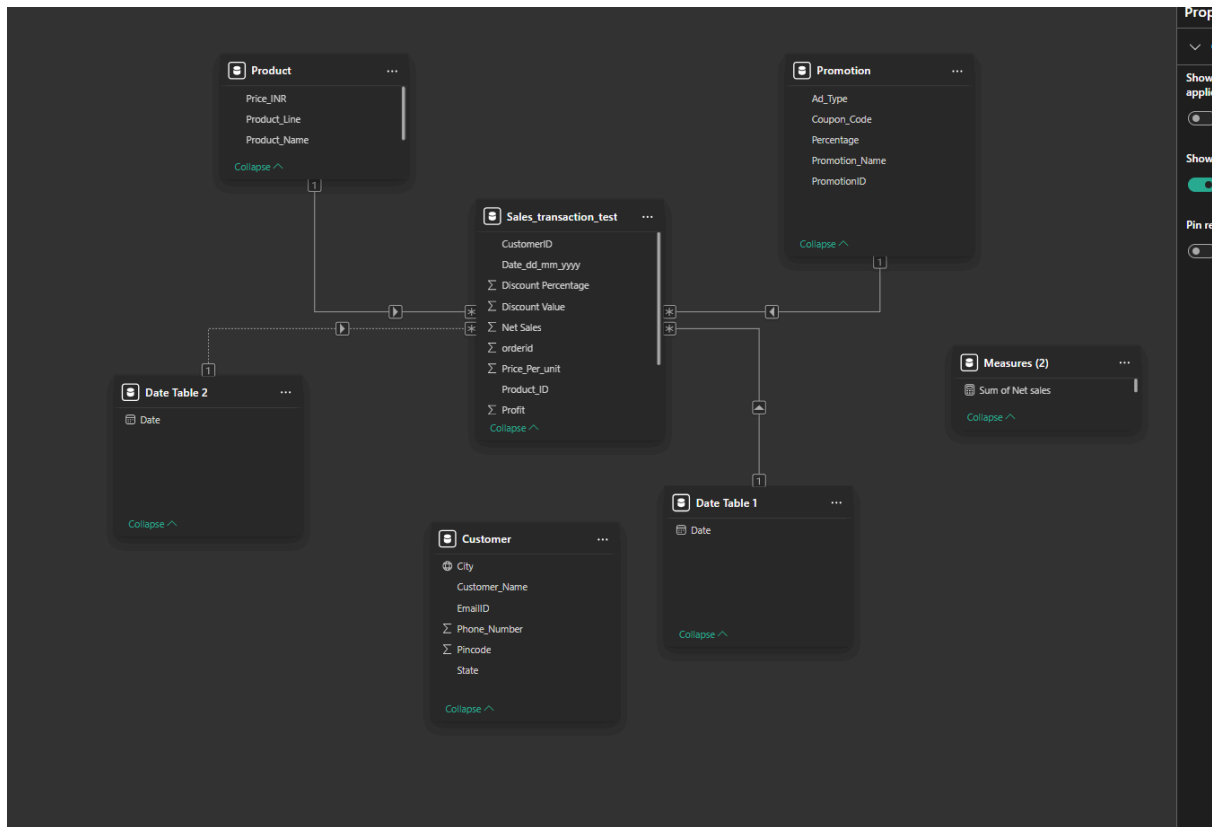
- One-to-Many Cardinality
- Active Relationship

This allowed customer-level analysis such as city-wise sales distribution and customer-based filtering.



Additional relationships were similarly created between the Sales Transactions table and the Product and Promotion tables using their respective IDs to maintain a centralized reporting structure.

The star schema design helped simplify the data model, improve filtering behavior, and support efficient dashboard performance during analysis.



Dashboard Development & KPI Visualization

After completing the data modeling process, the next step was to build an interactive dashboard in Power BI to visualize key business metrics and trends.

1. Profit Calculation

A new **Profit** column was created based on the business assumption that profit would be calculated as 10% of Net Sales.

This metric was later used for profitability analysis and product performance tracking across the dashboard.

2. Sales Trend Analysis

A time-series sales analysis was created to monitor how sales changed over different time periods.

The dashboard included trend visualizations for:

- daily sales,
- monthly performance,
- quarterly trends, and

- yearly sales movement.

This helped identify seasonal patterns and overall business growth trends over time.

3. Sales vs Profit Analysis

To analyze the relationship between sales and profitability, a scatter plot visualization was used.

The scatter plot helped compare how product sales translated into profit across different categories and transactions. This made it easier to identify products or regions with high sales but relatively lower profitability.

4. Average Discount Analysis

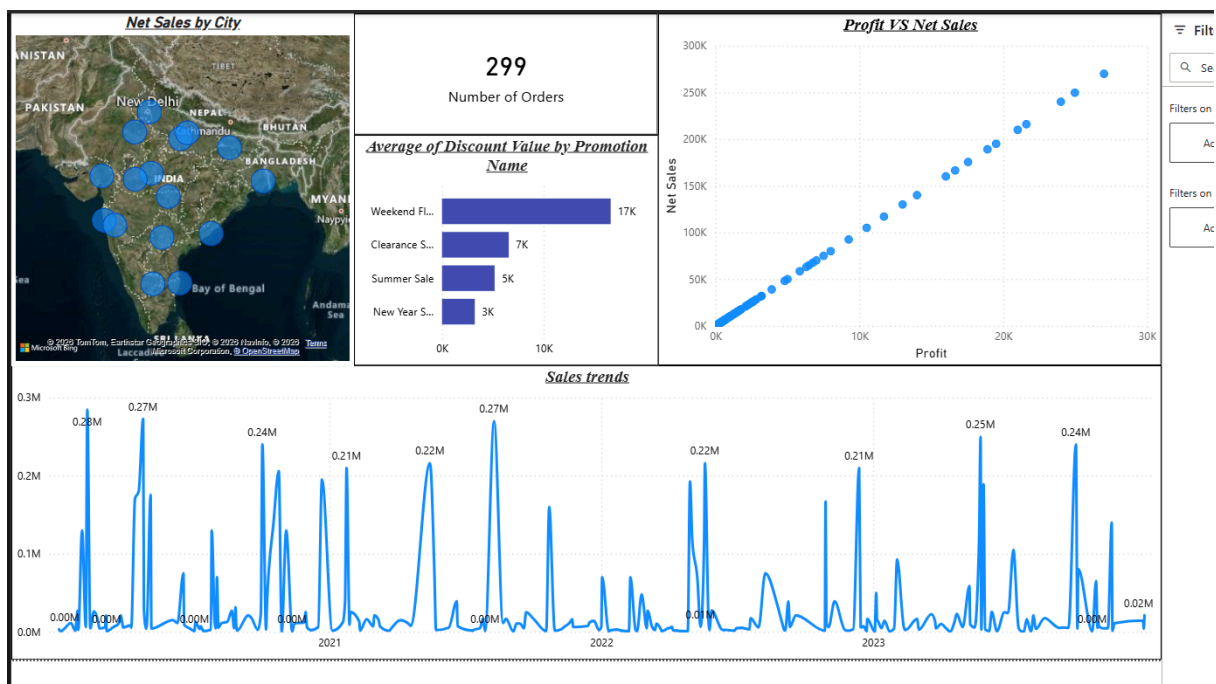
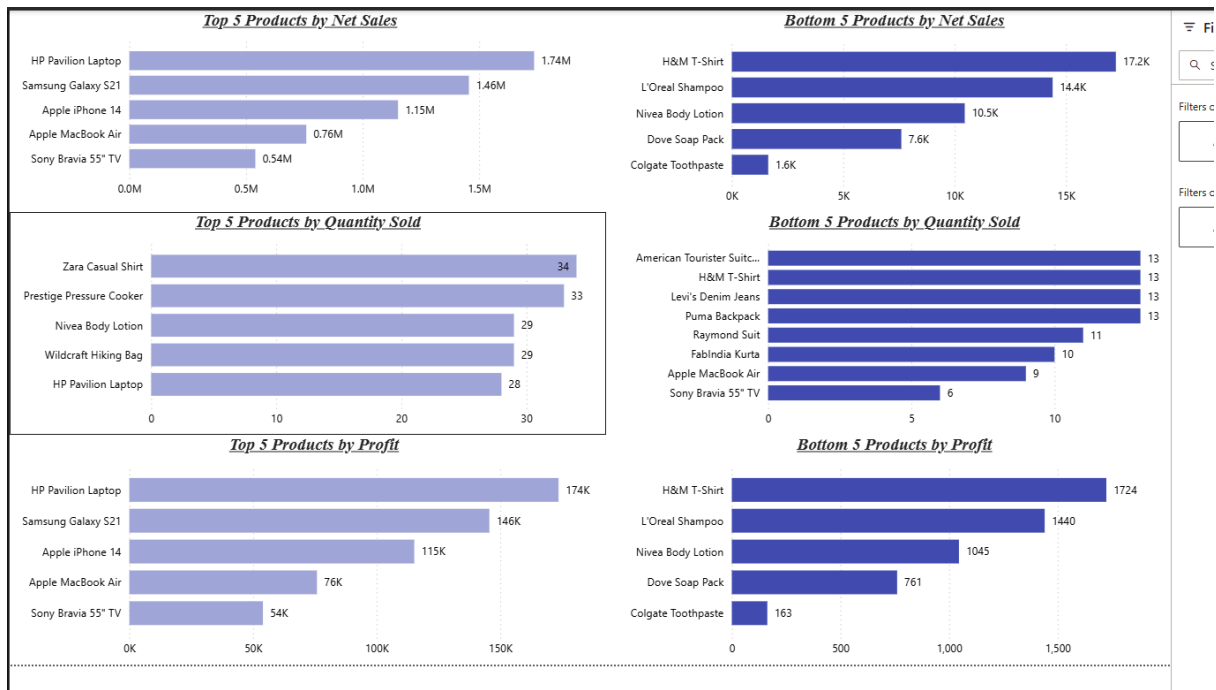
A separate visualization was created to analyze the average discount offered under different promotion categories.

This analysis helped evaluate discount strategies and understand how promotional campaigns were impacting sales performance.

5. Geographic Sales Analysis

A map visualization was used to display Net Sales distribution across different cities.

This helped identify high-performing regions and provided better visibility into geographical sales patterns across the business.



Comparative Period Analysis

One of the business requirements was to allow users to compare performance across two different time periods.

To achieve this, two separate date tables were used within the data model. The primary date table was connected through the active relationship and was used for standard reporting and filtering.

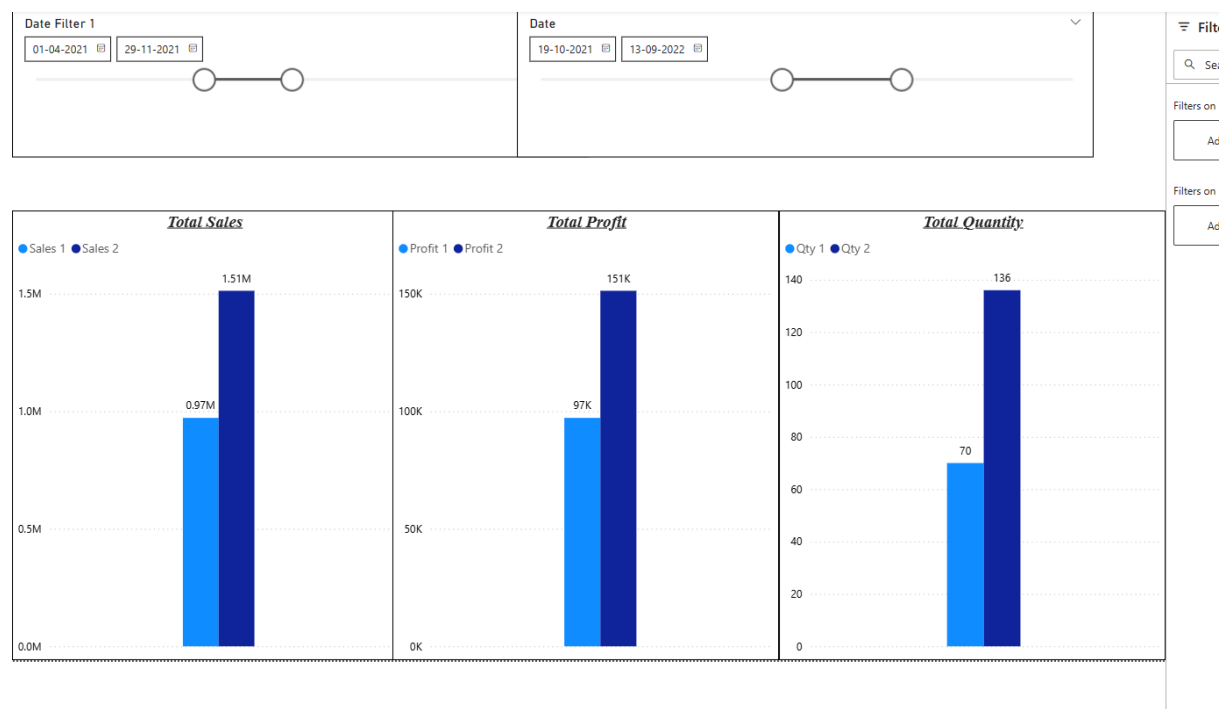
A second date table was created specifically for comparison purposes. Since only one active relationship can exist between the same date field and fact table, an inactive relationship was established between the secondary date table and the Sales Transactions table.

A DAX measure was then created using the `USERELATIONSHIP()` function to temporarily activate the secondary relationship during calculation.

```
Sum of Net Sales =
CALCULATE(
    SUM(Sales_transaction_test[Net Sales]),
    ALL('Date Table 1'),
    USERELATIONSHIP(
        'Date Table 2'[Date],
        Sales_transaction_test[Date_dd_mm_yyyy]
    )
)
```

This approach enabled users to select two different date ranges and compare business performance without affecting the primary reporting period.

The feature allowed side-by-side analysis of Net Sales across different time periods, helping users identify growth patterns, seasonal changes, and performance differences between selected periods.

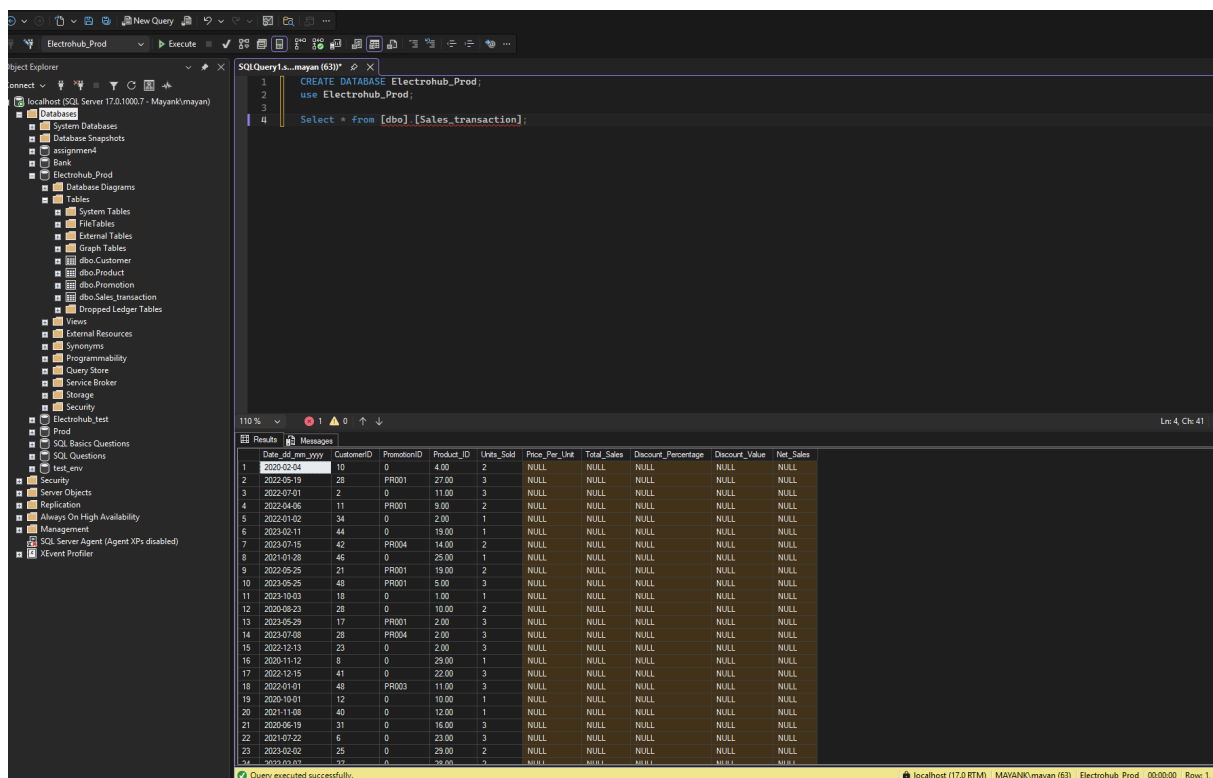


Transition from Test to Production Environment

Once the data model, calculations, relationships, and dashboard visuals were validated in the testing environment, the project was migrated to the production dataset.

During the development phase, a smaller test dataset containing approximately 300 sales transaction records was used to verify:

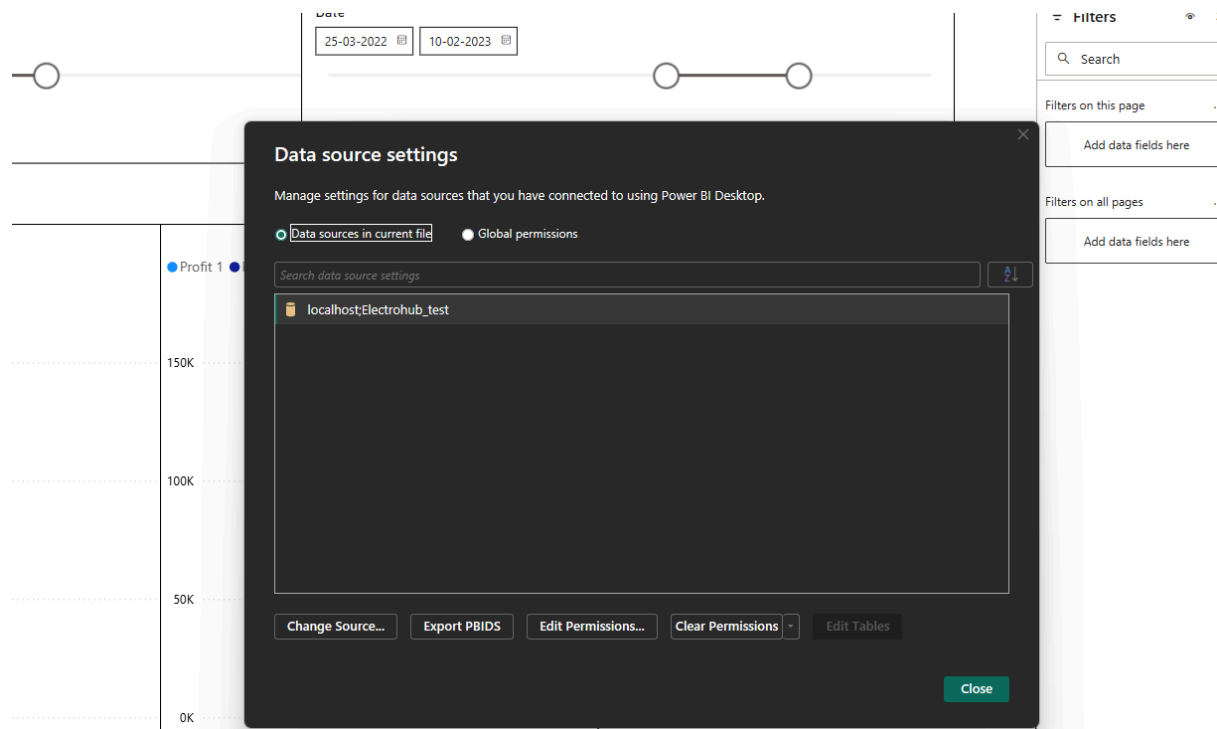
- data transformations,
- table relationships,
- KPI calculations,
- DAX measures,
- filter interactions, and
- dashboard functionality.



After successful validation, the data source was updated to the production environment, which contained the complete sales transaction dataset with more than 3,000 records.

The transition ensured that all calculations and visualizations behaved consistently when working with a larger volume of data. It also provided

confidence that the dashboard was ready for business reporting and analysis on the full dataset.



This staged development approach helped reduce errors during implementation and made troubleshooting easier during the dashboard development process.

